

Q1 Show that $\mathcal{M}_2$ is a vector space under the standard operations.

Let $A = [a_{ij}], B = [b_{ij}], C = [c_{ij}] \in \mathcal{M}_2$, for $i, j = 1, 2$ and let u, v be scalars.

$$A + B = [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}] \in \mathcal{M}_2$$

$$uA = u[a_{ij}] = [ua_{ij}] \in \mathcal{M}_2$$

$$A + B = [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}] = [b_{ij} + a_{ij}] = [b_{ij}] + [a_{ij}] = B + A$$

$$\begin{aligned} (A + B) + C &= ([a_{ij}] + [b_{ij}]) + [c_{ij}] = [a_{ij} + b_{ij}] + [c_{ij}] \\ &= [a_{ij} + b_{ij} + c_{ij}] = [a_{ij}] + [b_{ij} + c_{ij}] = A + (B + C) \end{aligned}$$

Let $O = [o_{ij}]$ be the zero vector of $\mathcal{M}_2$ s.t. $A + O = A = O + A$ for any matrix A . Then

$$\begin{aligned} A + O = A &\implies [a_{ij} + o_{ij}] = [a_{ij}] \implies a_{ij} + o_{ij} = a_{ij} \text{ for } i, j = 1, 2 \\ &\implies o_{ij} = 0 \text{ for } i, j = 1, 2 \end{aligned}$$

$$\therefore O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ is the zero vector of } \mathcal{M}_2.$$

Let $A^* = [a_{ij}^*]$ denote the negative of A s.t. $A + A^* = O = A^* + A$. Then

$$\begin{aligned} A + A^* = O &\implies [a_{ij}] + [a_{ij}^*] = O \implies [a_{ij} + a_{ij}^*] = O \\ &\implies a_{ij} + a_{ij}^* = 0 \text{ for } i, j = 1, 2. \\ &\implies a_{ij}^* = -a_{ij} \text{ for } i, j = 1, 2. \end{aligned}$$

$$\therefore A^* = [-a_{ij}] = -[a_{ij}] = -A \text{ is the negative of } A.$$

$$u(A + B) = u[a_{ij} + b_{ij}] = [u(a_{ij} + b_{ij})] = [ua_{ij} + ub_{ij}] = [ua_{ij}] + [ub_{ij}] = u[a_{ij}] + u[b_{ij}] = uA + uB$$

$$(u + v)A = (u + v)[a_{ij}] = [(u + v)a_{ij}] = [ua_{ij} + va_{ij}] = [ua_{ij}] + [va_{ij}] = u[a_{ij}] + v[a_{ij}] = uA + vA$$